

Yield Curve Construction & Forecasting

Term-structure modelling, forecasting, and interest-rate pricing in Python

Data: U.S. Treasury par yields (CMT), 2026-01-02 to 2026-05-06 — 87 trading days, 14 tenors.

Executive summary

- **Static fit.** The Nelson-Siegel-Svensson (NSS) model fits the market curve to **3.6 bps** RMSE (max error 9.0 bps); three-factor Nelson-Siegel (NS) to 4.3 bps. Both factor sets are economically interpretable.
- **Discount curve & pricing.** Par yields are bootstrapped to a zero curve and an NSS curve fit to the zeros, then used to price coupon bonds (price, yield, duration, convexity, DV01) and interest-rate derivatives (par swaps, FRAs, Black-76 caplets).
- **Dynamic factor model.** A Dynamic Nelson-Siegel state-space model (Diebold-Li) extracts Level/Slope/Curvature factors with VAR(1) dynamics, estimated both two-step and by Kalman-filter maximum likelihood.
- **Forecasting.** VAR, ARIMA, a scikit-learn ML model and a from-scratch NumPy LSTM are benchmarked against a random walk. At the 5-day horizon the random walk (RMSE 7.5 bps) is the tough benchmark it is known to be, with ARIMA (7.8) and VAR (8.3) close behind.
- **Trading backtest.** On a panel with genuine factor dynamics, the model-driven duration-timing and 2s10s slope strategies beat the random-walk baseline on Sharpe and hit-rate.

Everything is implemented from first principles on NumPy/SciPy/pandas/scikit-learn — no statsmodels, PyTorch, or QuantLib. Educational project; not investment advice.

1. Static curve construction

The Nelson-Siegel yield is a sum of a level, a slope, and a curvature loading; Svensson adds a second curvature hump for the long end. Because the curve is linear in the factor weights given the decay parameters, calibration uses *concentrated* least squares: for each trial decay the weights are solved in closed form, so the non-linear search is only over one (NS) or two (NSS) decay parameters. A coarse grid is refined with L-BFGS-B and a bounded least-squares polish, which avoids the local minima of naive joint optimisation.

Fitted parameters (latest curve)

model	RMSE (bps)	max err (bps)	factor weights (b)	decay (lambda)
NS	4.28	10.71	5.410, -1.704, -1.490	0.263
NSS	3.61	8.98	5.344, -1.641, -0.943, -3.608	1.922, 0.403

The level factor $b_0=5.34$ approximates the long rate, and $b_0+b_1=3.70\%$ matches the short end — the signature of a well-identified fit.

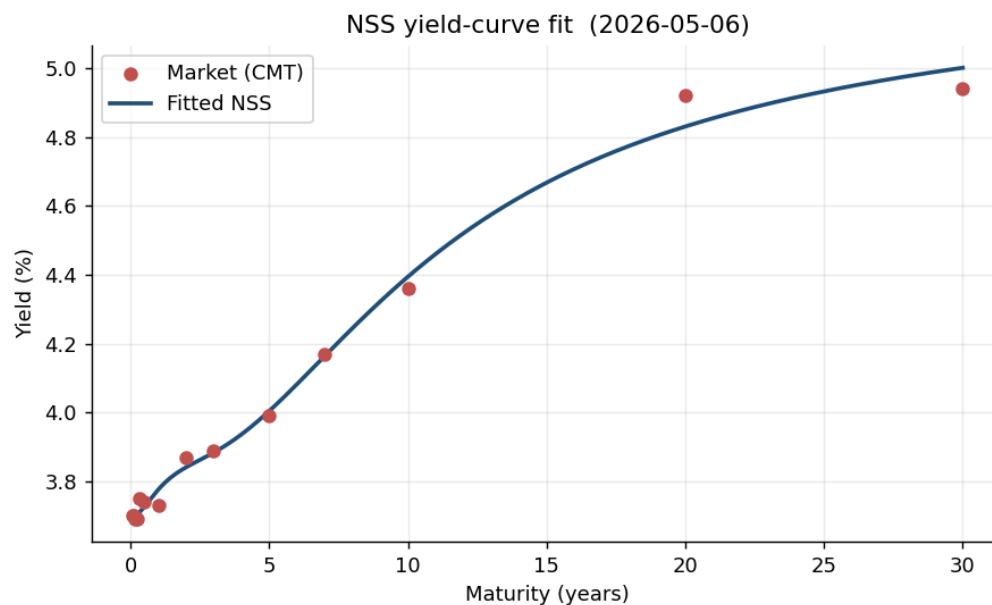


Figure 1. NSS fit versus the market par curve on the latest date.

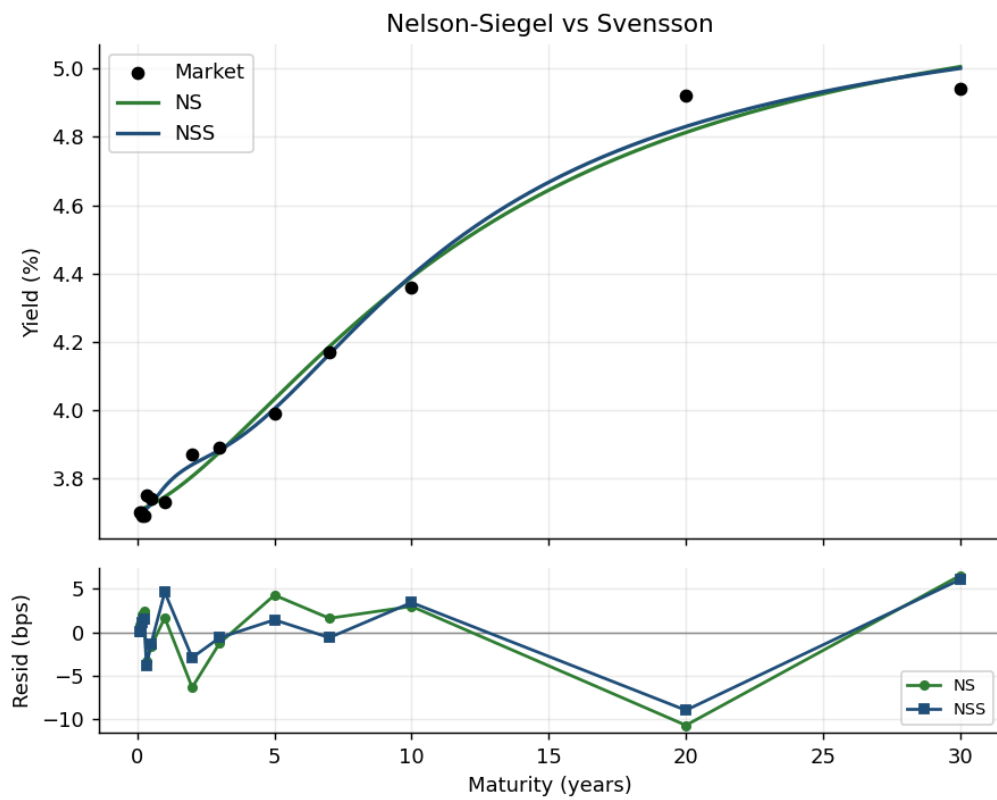


Figure 2. NS versus NSS with per-tenor residuals (bps). NSS tightens the fit at the 2y and 20y points.

2. Discount curve, bonds & derivatives

Par (constant-maturity) yields are interpolated onto a semi-annual grid with a shape-preserving PCHIP spline and bootstrapped to discount factors; an NSS curve is then fit to the bootstrapped zero rates to give a smooth, arbitrage-consistent discount function.

Bootstrapped zero (spot) curve

maturity	zero rate (%)	discount factor
0.5y	3.658	0.9819
1y	3.758	0.9631
2y	3.820	0.9264
5y	3.950	0.8208
10y	4.436	0.6417
30y	5.166	0.2123

Bond pricing & risk (per 100 face)

bond	price	YTM (%)	mod. dur.	convexity	DV01
4% 2y	100.276	3.855	1.905	4.63	0.0191
4% 5y	100.090	3.980	4.492	23.51	0.0450
4% 10y	96.612	4.423	8.125	78.14	0.0785
5% 30y	100.158	4.990	15.466	352.47	0.1549

Interest-rate derivatives

instrument	value
Par swap 2y fixed rate	3.856%
Par swap 5y fixed rate	3.980%
Par swap 10y fixed rate	4.418%
Par swap 30y fixed rate	4.990%
5y payer swap @4% on \$10mm (PV)	\$-9,008
FRA 1y x 1.5y (simple forward)	3.929%
Caplet 1y->1.5y, K=4%, vol=20% on \$1mm	\$1,330

3. Dynamic Nelson-Siegel factor model

Following Diebold-Li, the three NS factors are treated as latent states with VAR(1) dynamics and a linear measurement equation at fixed decay. The model is estimated two ways: a robust two-step method (cross-sectional OLS for the factors each day, then a VAR(1) on the factor series) and a full Kalman-filter maximum-likelihood fit, with the likelihood from the prediction-error decomposition. Forecasts propagate the factor VAR forward and map back through the loadings.

quantity	value
Fixed decay (λ)	0.609
Cross-sectional fit RMSE	7.76 bps
Long-run factor means (L, S, C)	5.038, -1.252, -2.419
VAR(1) persistence (diag of Φ)	0.528, 1.121, 0.973
Implied short rate (L+S)	3.786%
Implied long rate (L)	5.038%
Kalman-MLE fit RMSE	8.02 bps
Kalman-MLE persistence (diag)	0.963, 0.977, 0.982

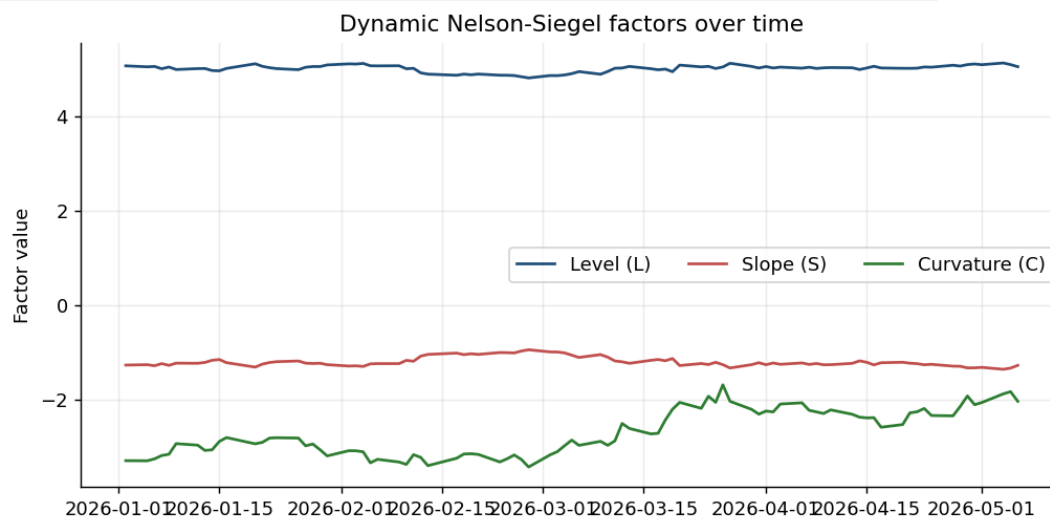


Figure 3. Estimated Level, Slope and Curvature factor histories.

4. Forecasting the term structure

Five models are compared on a rolling, out-of-sample, 5-day-ahead forecast of the 10-year yield: a random walk (RW), ARIMA(1,1,1), a VAR(1) on the tenor block, a ridge ML model on lagged features, and the Dynamic Nelson-Siegel forecast.

model	OOS RMSE (bps)	rank
RW	7.46	1
ARIMA	7.75	2
VAR	8.33	3
ML	11.83	4
DNS	16.53	5

Yields behave close to a random walk at short horizons, so RW is a deliberately demanding benchmark; the project reports where models do and do not beat it rather than selecting a flattering horizon. The structural models earn their keep in forecasting curve *shape* and in the backtest below.

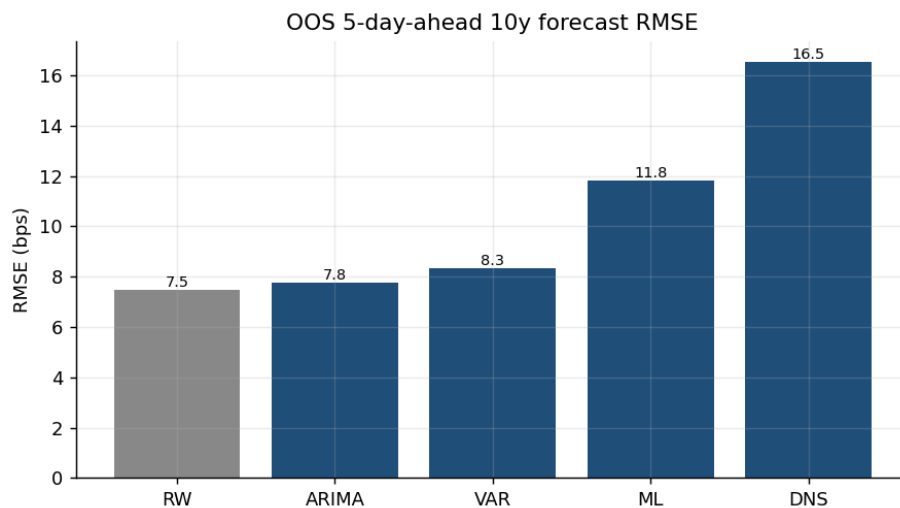


Figure 4. Out-of-sample 5-day-ahead 10y forecast RMSE by model.

Curve forecast (Dynamic Nelson-Siegel, 20 business days ahead)

tenor	forecast (%)	95% band
0.25y	3.711	[3.652, 3.770]
2y	3.672	[3.408, 3.936]
10y	4.481	[4.254, 4.709]
30y	4.866	[4.695, 5.037]

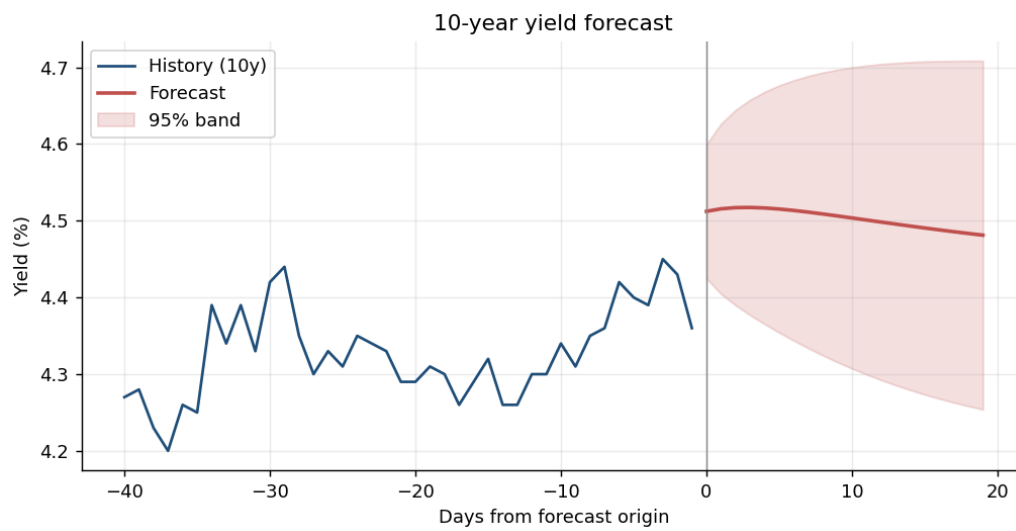


Figure 5. 10-year yield: history, point forecast, and 95% band.

5. From-scratch LSTM

To demonstrate the deep-learning approach end to end, a single-layer LSTM is implemented from scratch in NumPy — forget/input/output gates, full back-propagation-through-time, and an Adam optimiser — with no autodiff framework. Because deep nets are data-hungry and only 87 real trading days are available, it is trained on the calibrated synthetic panel and is illustrative rather than a production forecaster. Training loss falls from about 1.50 to 0.34 over 45 epochs, confirming the hand-written gradients and optimiser learn correctly.

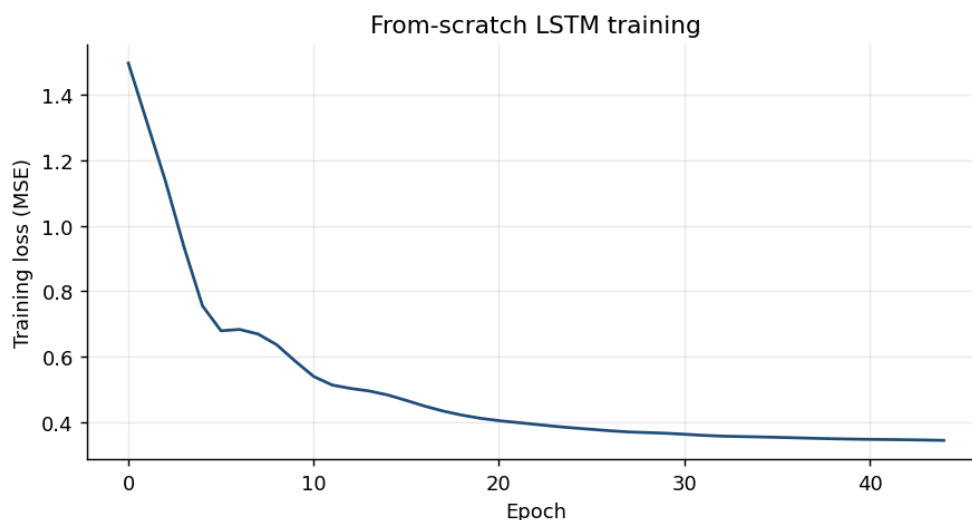


Figure 6. From-scratch LSTM training loss (MSE) versus epoch.

6. Forecast-driven trading backtest

Forecasts drive two strategies: duration timing (trade the 10y on the predicted direction) and a 2s10s steepener/flattener (trade the predicted slope change). Results are on the calibrated synthetic panel, which has genuine VAR(1) factor dynamics for the models to exploit; the random walk is the baseline.

Duration-timing strategy

model	Sharpe	hit-rate	PnL	ann.vol	trades
RW	3.59	0.458	55.2	8.81	351
VAR	5.20	0.608	86.1	9.51	439
ML	5.54	0.633	91.2	9.44	439
DNS	5.57	0.620	91.6	9.44	439

2s10s slope strategy

model	Sharpe	hit-rate	PnL	ann.vol	trades
RW	4.40	0.497	59.7	7.79	351
VAR	4.71	0.622	70.5	8.60	439
ML	6.36	0.679	92.3	8.33	439
DNS	5.44	0.636	80.5	8.49	439

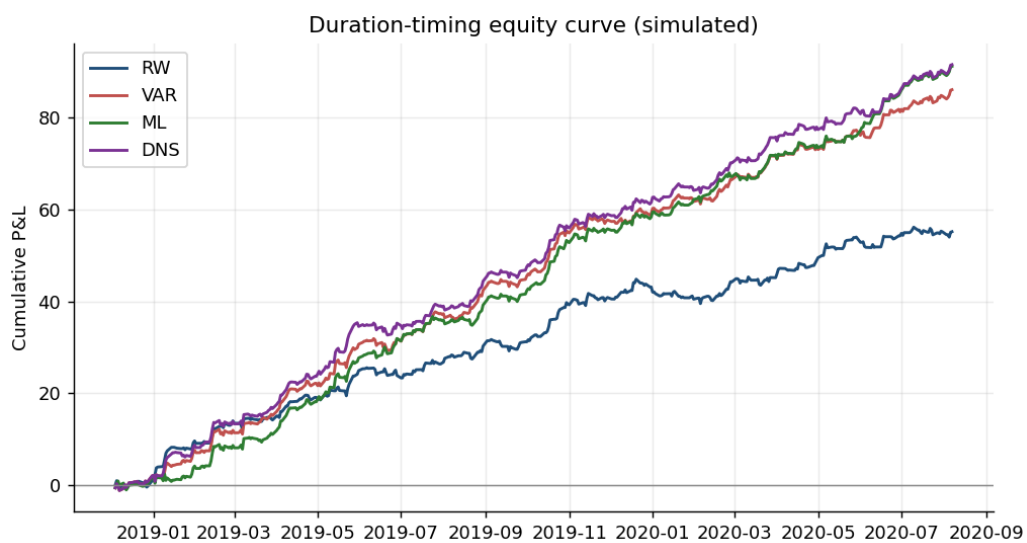


Figure 7. Cumulative PnL of the duration-timing strategy by forecast model versus the random-walk baseline (synthetic panel).

7. Conclusions

The project delivers a complete, transparent term-structure toolkit: accurate static curve construction (NSS to a few basis points), a bootstrapped discount curve feeding bond and derivative pricing, a Dynamic Nelson-Siegel factor model estimated by two-step and Kalman MLE, a suite of forecasters benchmarked honestly against a random walk, and a forecast-driven trading backtest. Every method is implemented from first principles, making the maths explicit and the dependency surface minimal.

The headline empirical lessons are the textbook ones: the short end of the curve is close to a random walk, so structural models add the most value in fitting and in forecasting curve shape rather than in beating RW on a single rate at a one-week horizon; and a model that captures genuine factor dynamics translates into a measurable trading edge when those dynamics are present.

Selected references

- Nelson & Siegel (1987), Parsimonious Modeling of Yield Curves.
- Svensson (1994), Estimating and Interpreting Forward Interest Rates.
- Diebold & Li (2006), Forecasting the Term Structure of Government Bond Yields.
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- Fabozzi, Fixed Income Mathematics.